

Vikram Dubois

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 73% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, C/C++, Python, FreeRTOS, PCB Debugging, MQTT

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — Tesla (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 46%.
- Led a squad of 8 engineers delivering scalable microservices processing 34M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 66%.

Associate Embedded Systems & IoT Firmware Engineer — STMicroelectronics (2019 - 2023)

- Designed and maintained C/C++ data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using Python saving \$11k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, C/C++, Python, FreeRTOS]

Engineered end-to-end production solution leveraging I2C/SPI/UART and C/C++ with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, FreeRTOS, PCB Debugging, MQTT]

Architected high-resilience system integrating Python with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

Vellore Institute of Technology (VIT) — 2015 - 2019 (Score: 3.78 / 4.0)